Final Project Code Appendix

0. Imports and Load Packages

```
library(tidyverse)
library(lmtest)
library(sandwich)
library(scales)
```

1. Read and clean data

```
salary <- read.table(
  "salary.txt",
  header = TRUE,
  sep = "",
  stringsAsFactors = FALSE,
  na.strings = "NA"
)

salary <- salary %>%
  mutate(
    sex = factor(sex, levels = c("M", "F")),
    deg = factor(deg),
    field = factor(field),
    rank = factor(rank, levels = c("Assist", "Assoc", "Full")),
    admin = factor(admin, levels = c(0, 1)),
    startyr = as.integer(startyr),
    year = as.integer(year),
    yrdeg = as.integer(yrdeg),
    salary = as.numeric(salary),
    experience = year - startyr
  )

# checks for problematic values (ensured no negative experience)
# salary %>%
  # filter(experience < 0) %>%
  # print()

salary_clean <- salary %>%
  filter(experience >= 0, !is.na(rank))
```

2. Descriptive summaries

```
salary_clean %>%
  count(sex, rank, field)
```

```
##    sex   rank field    n
```

```
## 1    M Assist  Arts  349
## 2    M Assist Other 1614
## 3    M Assist  Prof  625
## 4    M  Assoc  Arts  929
## 5    M  Assoc Other 3197
## 6    M  Assoc  Prof  938
## 7    M   Full  Arts  759
## 8    M   Full Other 5594
## 9    M   Full  Prof 1857
## 10   F Assist  Arts  319
## 11   F Assist Other 1012
## 12   F Assist  Prof  129
## 13   F  Assoc  Arts  300
## 14   F  Assoc Other 1032
## 15   F  Assoc  Prof  133
## 16   F   Full  Arts  183
## 17   F   Full Other  691
## 18   F   Full  Prof  127
```

```r
salary_clean %>%
  summarise(
    n_records = n(),
    n_people = n_distinct(id),
    min_year = min(year, na.rm = TRUE),
    max_year = max(year, na.rm = TRUE)
  ) %>%
  print()
```

```
##   n_records n_people min_year max_year
## 1     19788     1597       76       95
```

3. Q1: Starting salary discrimination

```r
start_salary <- salary_clean %>%
  filter(year == startyr)

start_salary %>%
  summarise(
    n_records = n(),
    n_people = n_distinct(id)
  ) %>%
  print()
```

```
##   n_records n_people
## 1      1107     1107
```

```r
cat("\nRank distribution at hire:\n")
```

```
##
## Rank distribution at hire:
```

```r
start_salary %>%
  count(rank, sex) %>%
  arrange(rank, sex)
```

```
##     rank sex   n
```

```
## 1 Assist   M 501
## 2 Assist   F 301
## 3  Assoc   M 138
## 4  Assoc   F  38
## 5   Full   M 114
## 6   Full   F  15
```

```r
cat("\nStarting salary summaries by sex:\n")
```

```
##
## Starting salary summaries by sex:
```

```r
q1_summary <- start_salary %>%
  group_by(sex) %>%
  summarise(
    n = n(),
    mean_salary = mean(salary, na.rm = TRUE),
    median_salary = median(salary, na.rm = TRUE),
    sd_salary = sd(salary, na.rm = TRUE),
    iqr_salary = IQR(salary, na.rm = TRUE)
  )
print(q1_summary)
```

```
## # A tibble: 2 x 6
##   sex       n mean_salary median_salary sd_salary iqr_salary
##   <fct> <int>       <dbl>         <dbl>     <dbl>      <dbl>
## 1 M       753       4016.         3719      1933.       2076
## 2 F       354       3600.         3502.     1348.       1477
```

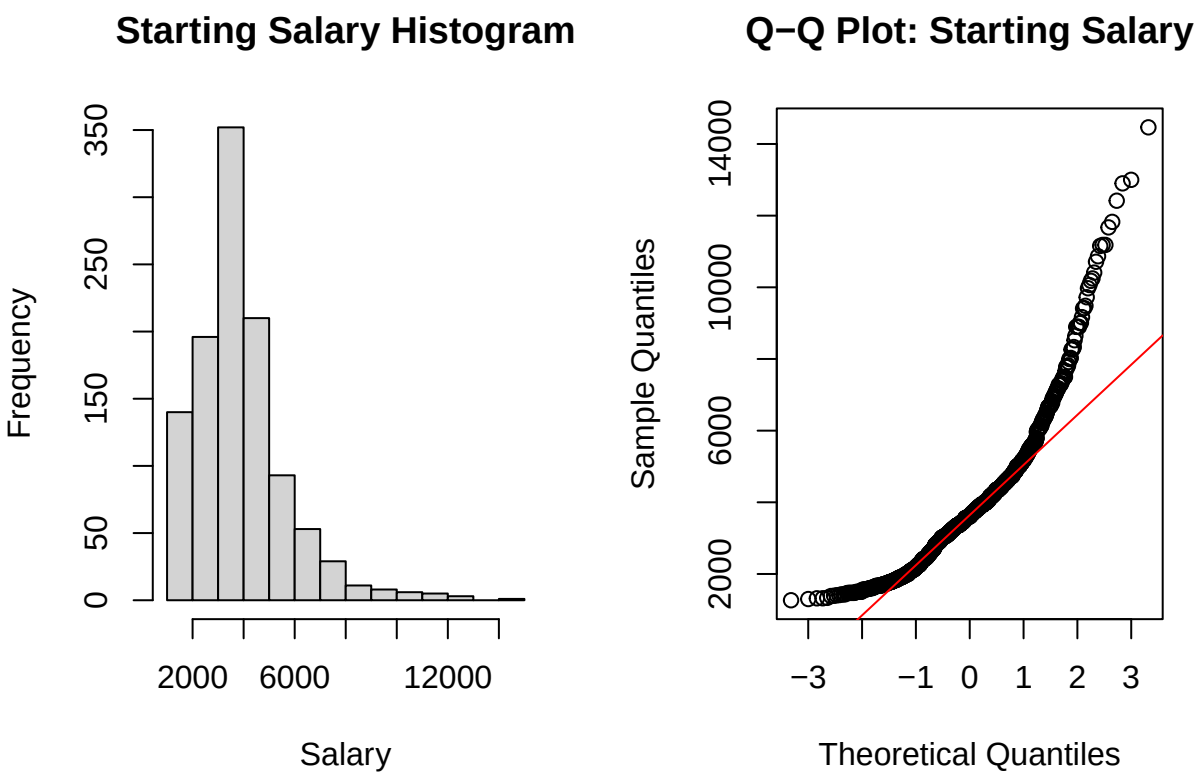
```r
cat("\nStarting salary summaries by sex and field:\n")
```

```
##
## Starting salary summaries by sex and field:
```

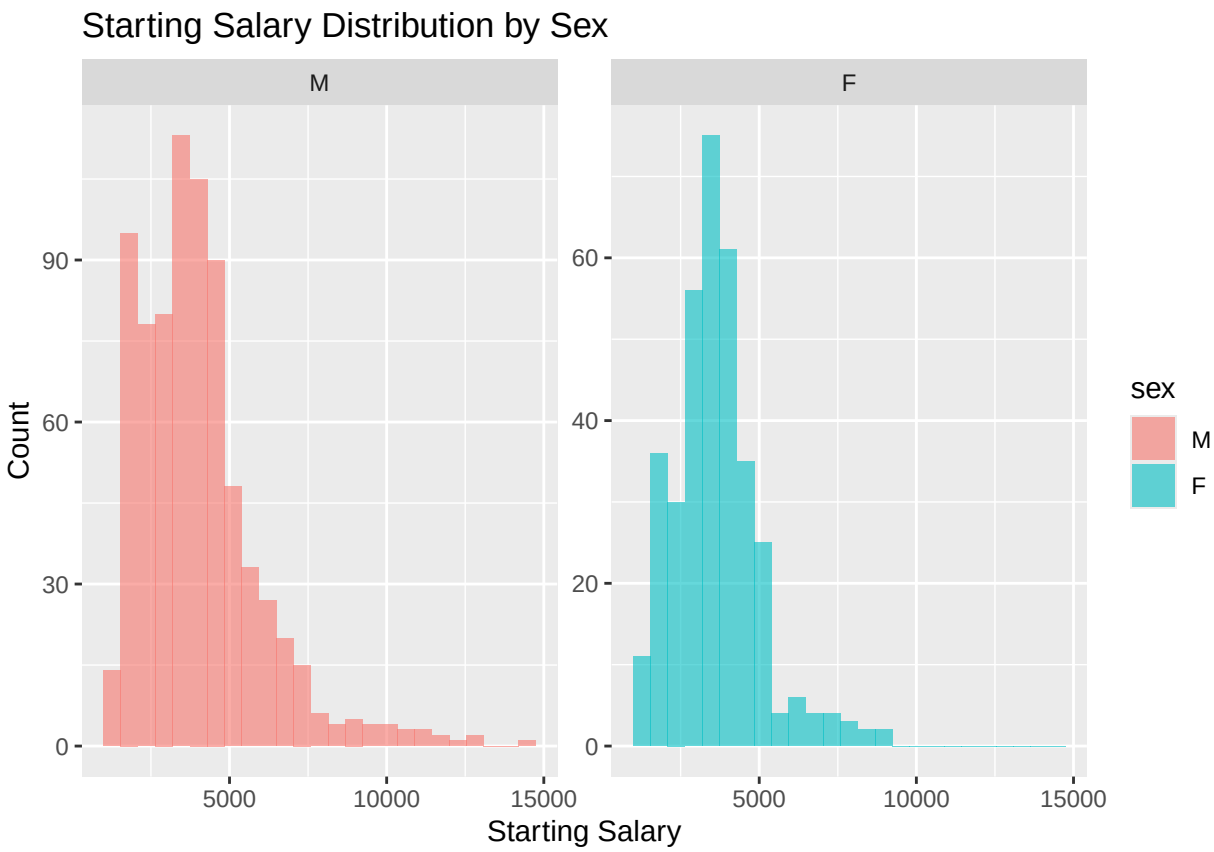
```r
start_salary %>%
  group_by(field, sex) %>%
  summarise(
    n = n(),
    mean_salary = mean(salary, na.rm = TRUE),
    median_salary = median(salary, na.rm = TRUE),
    .groups = "drop"
  )
```

```
## # A tibble: 6 x 5
##   field sex       n mean_salary median_salary
##   <fct> <fct> <int>       <dbl>         <dbl>
## 1 Arts  M        76       3137.         3029
## 2 Arts  F        71       3083.         3271
## 3 Other M       480       4048.         3721
## 4 Other F       247       3610.         3500
## 5 Prof  M       197       4276.         4043
## 6 Prof  F        36       4553.         4678.
```

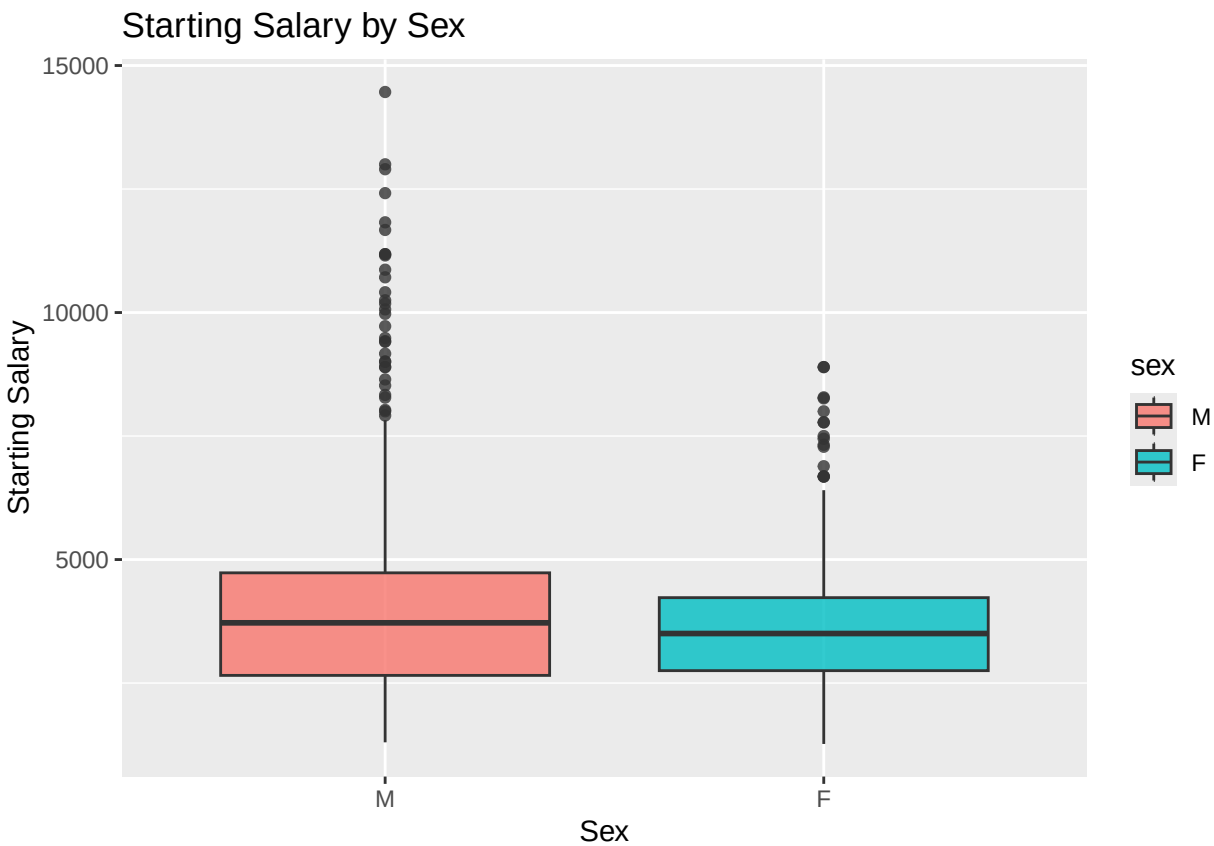
```r
par(mfrow = c(1, 2))
hist(start_salary$salary, main = "Starting Salary Histogram", xlab = "Salary")
qqnorm(start_salary$salary, main = "Q-Q Plot: Starting Salary")
qqline(start_salary$salary, col = "red")
```




```r
par(mfrow = c(1, 1))

ggplot(start_salary, aes(x = salary, fill = sex)) +
  geom_histogram(alpha = 0.6, bins = 25, position = "identity") +
  facet_wrap(~ sex, scales = "free_y") +
  labs(
    title = "Starting Salary Distribution by Sex",
    x = "Starting Salary",
    y = "Count"
  )
```




```r
ggplot(start_salary, aes(x = sex, y = salary, fill = sex)) +
  geom_boxplot(alpha = 0.8) +
  labs(
    title = "Starting Salary by Sex",
    x = "Sex",
    y = "Starting Salary"
  )
```




```r
ggplot(start_salary, aes(x = factor(startyr), y = salary, fill = sex)) +
  geom_boxplot(outlier.alpha = 0.3) +
  labs(
    title = "Starting Salary by Hire Year and Sex",
    x = "Hire Year",
    y = "Starting Salary"
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```r
cat("\nQ1 Welch two-sample t-test:\n")
```

```
##
## Q1 Welch two-sample t-test:
```

```r
q1_ttest <- t.test(salary ~ sex, data = start_salary)
print(q1_ttest)
```

```
##
##  Welch Two Sample t-test
##
## data:  salary by sex
## t = 4.1326, df = 949.26, p-value = 3.904e-05
## alternative hypothesis: true difference in means between group M and group F is not equal to 0
## 95 percent confidence interval:
##  218.0416 612.4033
## sample estimates:
## mean in group M mean in group F
##        4015.545        3600.323
```

```r
q1_lm <- lm(salary ~ sex + field + yrdeg + startyr, data = start_salary)
cat("\nQ1 linear regression on raw salary:\n")
```

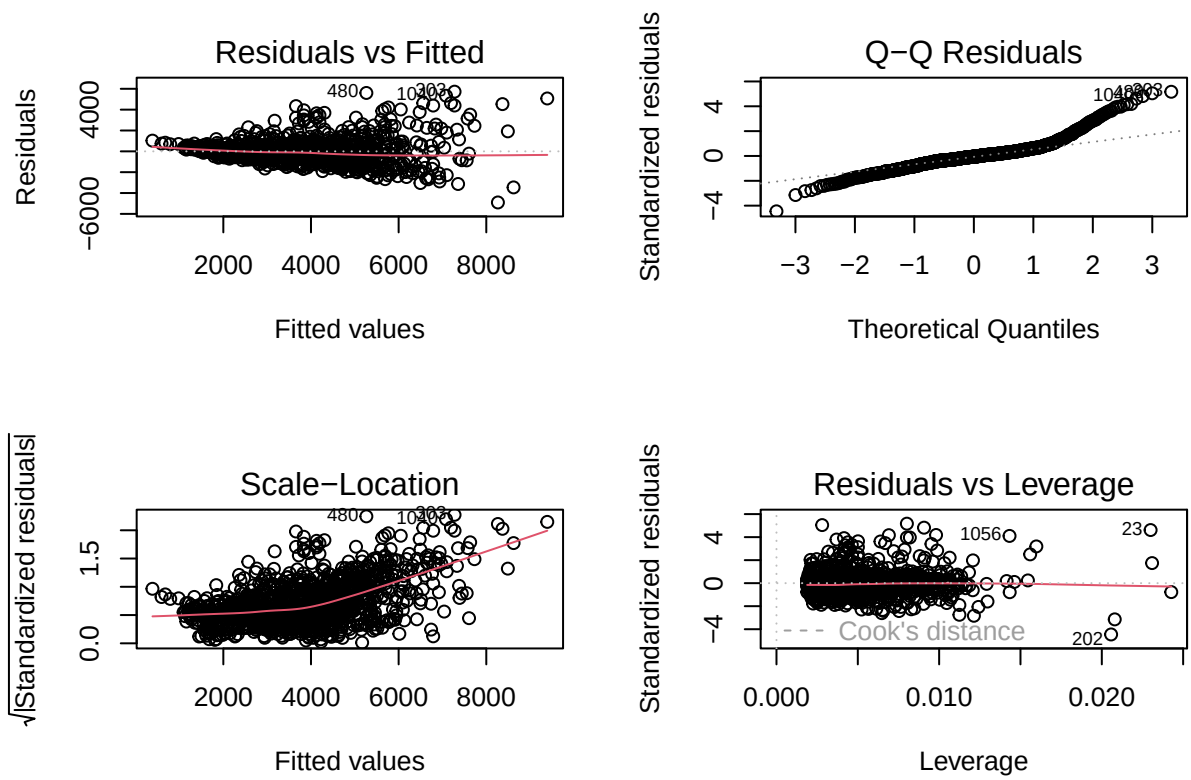
```
##
## Q1 linear regression on raw salary:
```

```r
summary(q1_lm)
```

```
##
## Call:
```

```
## lm(formula = salary ~ sex + field + yrdeg + startyr, data = start_salary)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4903.7  -524.8   -64.5   373.0  5723.9
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -12782.653    524.223 -24.384  < 2e-16 ***
## sexF          -291.212     75.145  -3.875 0.000113 ***
## fieldOther     483.298    101.327   4.770 2.09e-06 ***
## fieldProf     1122.489    119.575   9.387  < 2e-16 ***
## yrdeg         -113.564      5.033 -22.562  < 2e-16 ***
## startyr        292.127      7.297  40.032  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1112 on 1101 degrees of freedom
## Multiple R-squared:  0.6103, Adjusted R-squared:  0.6085
## F-statistic: 344.9 on 5 and 1101 DF,  p-value: < 2.2e-16
```

```r
par(mfrow = c(2, 2))
plot(q1_lm)
```




```r
par(mfrow = c(1, 1))

cat("\nQ1 regression with heteroskedasticity-robust SE:\n")
```

```
##
## Q1 regression with heteroskedasticity-robust SE:
```

```r
coeftest(q1_lm, vcov = vcovHC(q1_lm, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                Estimate  Std. Error  t value  Pr(>|t|)
## (Intercept) -12782.6525    475.8501 -26.8628 < 2.2e-16 ***
## sexF          -291.2120     67.2647  -4.3293 1.632e-05 ***
## fieldOther     483.2976     64.3585   7.5095 1.224e-13 ***
## fieldProf     1122.4886     92.0407  12.1956 < 2.2e-16 ***
## yrdeg         -113.5642      9.0566 -12.5394 < 2.2e-16 ***
## startyr        292.1267     11.4246  25.5699 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Log salary model
start_salary <- start_salary %>%
  mutate(log_salary = log(salary))

q1_lm_log <- lm(log_salary ~ sex + field + yrdeg + startyr, data = start_salary)
cat("\nQ1 linear regression on log salary:\n")
```

```
##
## Q1 linear regression on log salary:
```

```r
summary(q1_lm_log)
```

```
##
## Call:
## lm(formula = log_salary ~ sex + field + yrdeg + startyr, data = start_salary)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.99931 -0.14216 -0.00468  0.13354  0.79366
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.308369   0.109199  30.297  < 2e-16 ***
## sexF        -0.073436   0.015653  -4.691 3.05e-06 ***
## fieldOther   0.132527   0.021107   6.279 4.90e-10 ***
## fieldProf    0.278076   0.024908  11.164  < 2e-16 ***
## yrdeg       -0.022434   0.001049 -21.396  < 2e-16 ***
## startyr      0.075468   0.001520  49.648  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2316 on 1101 degrees of freedom
## Multiple R-squared:  0.7096, Adjusted R-squared:  0.7083
## F-statistic: 538.1 on 5 and 1101 DF,  p-value: < 2.2e-16
```

```r
cat("\nQ1 log-salary regression with robust SE:\n")
```

```
##
## Q1 log-salary regression with robust SE:
```

```r
coeftest(q1_lm_log, vcov = vcovHC(q1_lm_log, type = "HC1"))
```

```
##
## t test of coefficients:
##
##               Estimate Std. Error  t value  Pr(>|t|)
## (Intercept)  3.3083687  0.1052231  31.4415 < 2.2e-16 ***
## sexF        -0.0734358  0.0147648  -4.9737 7.615e-07 ***
## fieldOther   0.1325275  0.0150057   8.8318 < 2.2e-16 ***
## fieldProf    0.2780760  0.0203659  13.6540 < 2.2e-16 ***
## yrdeg       -0.0224342  0.0014548 -15.4211 < 2.2e-16 ***
## startyr      0.0754680  0.0018171  41.5316 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
q1_log_coef <- coef(q1_lm_log)["sexF"]
cat("\nApproximate percent difference for women in starting salary:\n")
```

```
##
## Approximate percent difference for women in starting salary:
```

```r
cat(round((exp(q1_log_coef) - 1) * 100, 2), "%\n")
```

```
## -7.08 %
```

4. Q2: Wage difference at promotion

```r
promotion_data <- salary_clean %>%
  arrange(id, year) %>%
  group_by(id) %>%
  mutate(
    prev_rank = lag(rank),
    prev_salary = lag(salary),
    prev_year = lag(year)
  ) %>%
  ungroup() %>%
  filter(prev_rank == "Assoc", rank == "Full")

promotion_data %>%
  summarise(
    n_promotions = n(),
    n_people = n_distinct(id)
  ) %>%
  print()
```

```
## # A tibble: 1 x 2
##   n_promotions n_people
##          <int>    <int>
## 1          547      547
```

```r
promotion_data <- promotion_data %>%
  group_by(id) %>%
  slice(1) %>%
  ungroup()
```

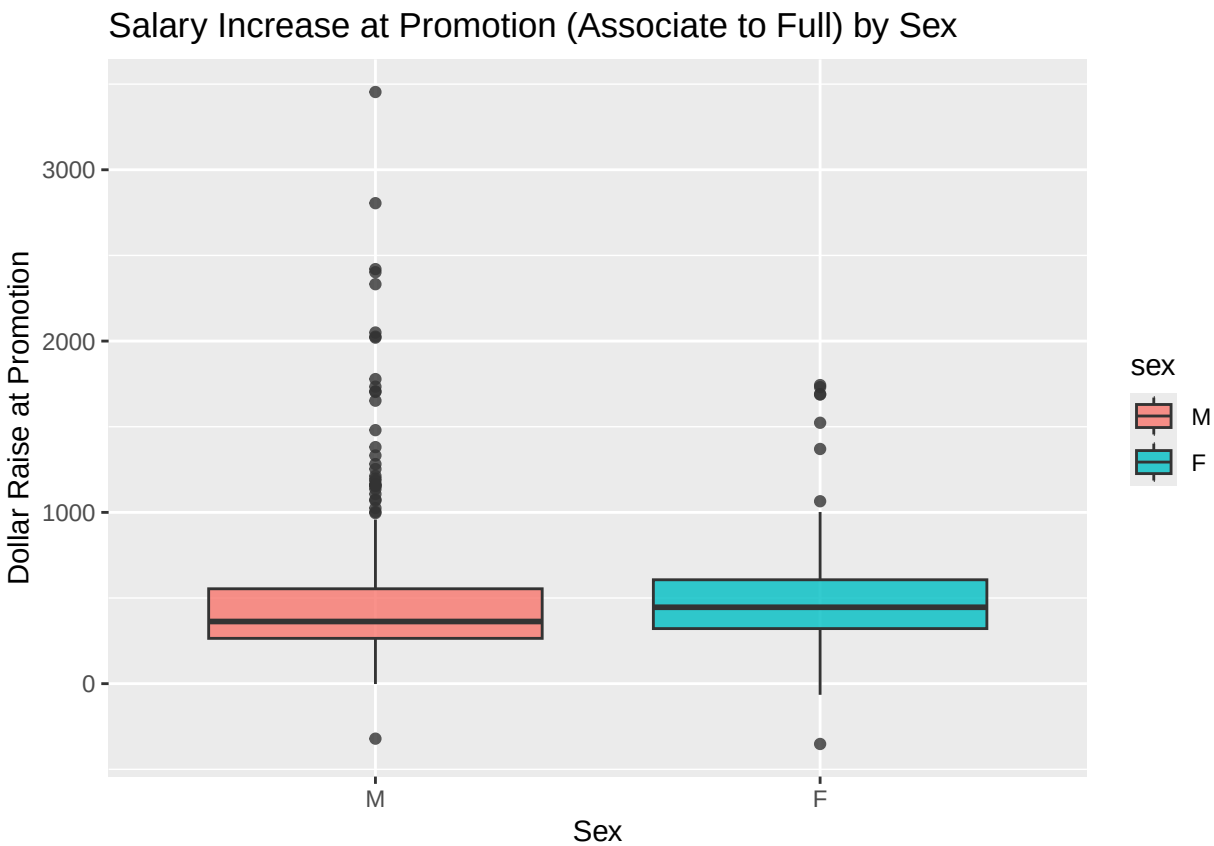
```r
promotion_data <- promotion_data %>%
  mutate(
    raise_amount = salary - prev_salary,
    raise_pct = (salary - prev_salary) / prev_salary
  )

cat("\nPromotion raise summaries by sex:\n")
```

```
##
## Promotion raise summaries by sex:
```

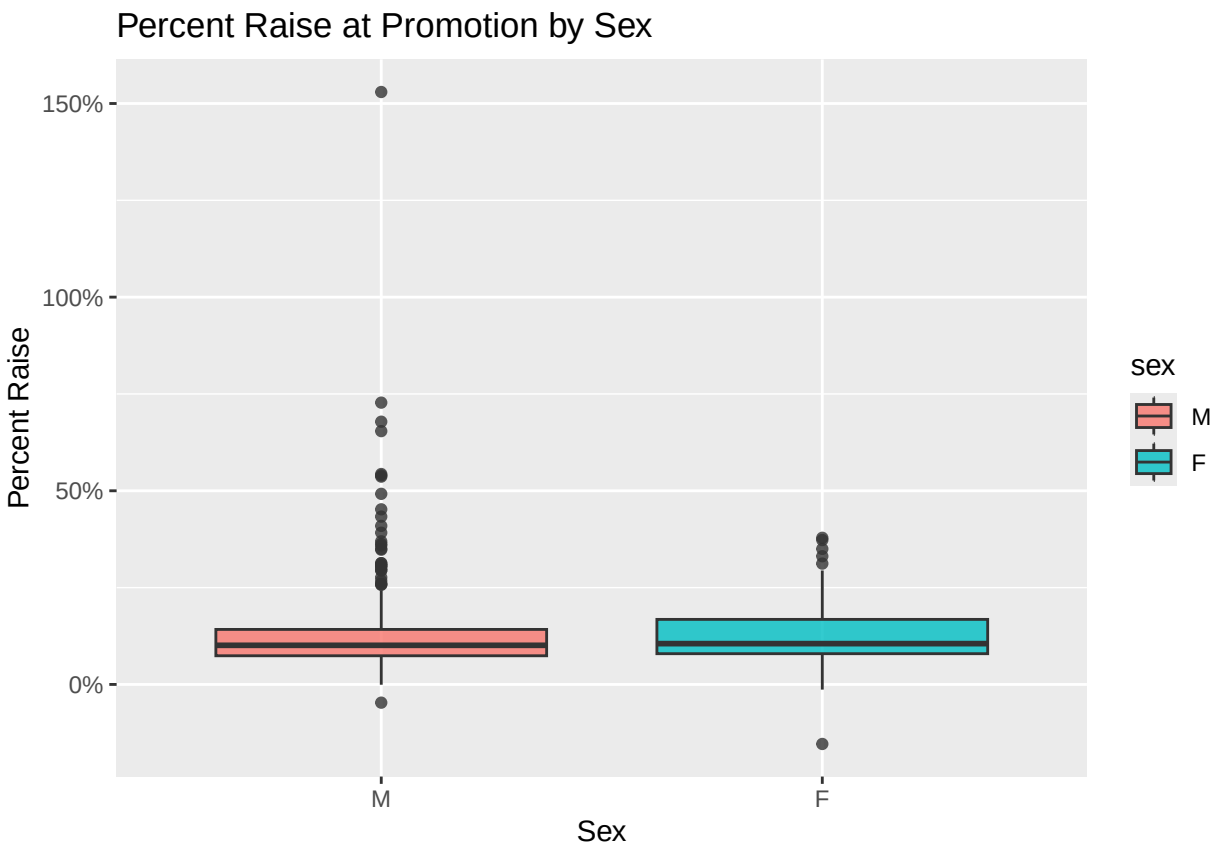
```r
promotion_data %>%
  group_by(sex) %>%
  summarise(
    n = n(),
    mean_raise = mean(raise_amount, na.rm = TRUE),
    median_raise = median(raise_amount, na.rm = TRUE),
    mean_pct_raise = mean(raise_pct, na.rm = TRUE),
    median_pct_raise = median(raise_pct, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  print()
```

```
## # A tibble: 2 x 6
##   sex       n mean_raise median_raise mean_pct_raise median_pct_raise
##   <fct> <int>      <dbl>        <dbl>          <dbl>            <dbl>
## 1 M       447       470.          363          0.124            0.101
## 2 F       100       517.          446          0.129            0.105
```

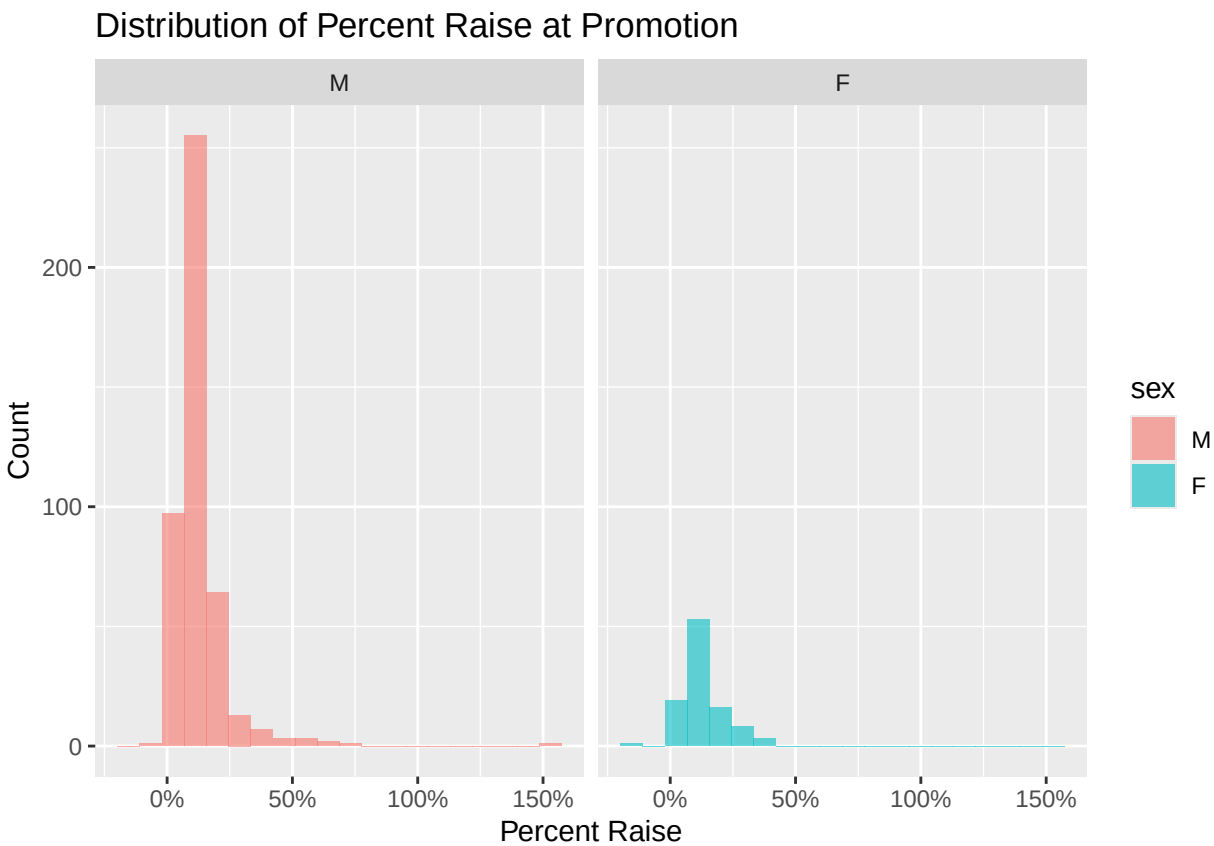
```r
ggplot(promotion_data, aes(x = sex, y = raise_amount, fill = sex)) +
  geom_boxplot(alpha = 0.8) +
  labs(
    title = "Salary Increase at Promotion (Associate to Full) by Sex",
    x = "Sex",
    y = "Dollar Raise at Promotion"
  )
```




```r
ggplot(promotion_data, aes(x = sex, y = raise_pct, fill = sex)) +
  geom_boxplot(alpha = 0.8) +
  scale_y_continuous(labels = percent_format(accuracy = 1)) +
  labs(
    title = "Percent Raise at Promotion by Sex",
    x = "Sex",
    y = "Percent Raise"
  )
```




```r
ggplot(promotion_data, aes(x = raise_pct, fill = sex)) +
  geom_histogram(alpha = 0.6, bins = 20, position = "identity") +
  facet_wrap(~ sex) +
  scale_x_continuous(labels = percent_format(accuracy = 1)) +
  labs(
    title = "Distribution of Percent Raise at Promotion",
    x = "Percent Raise",
    y = "Count"
  )
```




```r
cat("\nQ2 Welch t-test for dollar raise:\n")
```

```
##
## Q2 Welch t-test for dollar raise:
```

```r
print(t.test(raise_amount ~ sex, data = promotion_data))
```

```
##
##  Welch Two Sample t-test
##
## data:  raise_amount by sex
## t = -1.1385, df = 155.45, p-value = 0.2566
## alternative hypothesis: true difference in means between group M and group F is not equal to 0
## 95 percent confidence interval:
##  -127.73859   34.32727
## sample estimates:
## mean in group M mean in group F
##        469.9813        516.6870
```

```r
cat("\nQ2 Welch t-test for percent raise:\n")
```

```
##
## Q2 Welch t-test for percent raise:
```

```r
print(t.test(raise_pct ~ sex, data = promotion_data))
```

```
##
##  Welch Two Sample t-test
##
## data:  raise_pct by sex
```

```
## t = -0.45167, df = 182.87, p-value = 0.652
## alternative hypothesis: true difference in means between group M and group F is not equal to 0
## 95 percent confidence interval:
##  -0.02508576  0.01573983
## sample estimates:
## mean in group M mean in group F
##        0.124312        0.128985
```

```r
cat("\nQ2 Wilcoxon test for percent raise:\n")
```

```
##
## Q2 Wilcoxon test for percent raise:
```

```r
print(wilcox.test(raise_pct ~ sex, data = promotion_data))
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  raise_pct by sex
## W = 20318, p-value = 0.155
## alternative hypothesis: true location shift is not equal to 0
```

```r
q2_lm_raise <- lm(
  raise_amount ~ sex + field + yrdeg + prev_salary,
  data = promotion_data
)
cat("\nQ2 regression for dollar raise:\n")
```

```
##
## Q2 regression for dollar raise:
```

```r
summary(q2_lm_raise)
```

```
##
## Call:
## lm(formula = raise_amount ~ sex + field + yrdeg + prev_salary,
##     data = promotion_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1052.51  -191.06   -77.61    83.92  3059.16
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -6.858e+02  2.296e+02  -2.987  0.00295 **
## sexF         2.586e+01  4.282e+01   0.604  0.54612
## fieldOther  -8.941e+01  4.768e+01  -1.875  0.06128 .
## fieldProf    1.250e+00  5.730e+01   0.022  0.98261
## yrdeg        1.675e+01  3.629e+00   4.616 4.89e-06 ***
## prev_salary -1.156e-03  1.574e-02  -0.073  0.94150
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 377.3 on 541 degrees of freedom
## Multiple R-squared:  0.07412,    Adjusted R-squared:  0.06556
## F-statistic: 8.661 on 5 and 541 DF,  p-value: 6.54e-08
```

```r
cat("\nQ2 regression for dollar raise with robust SE:\n")
```

```
##
## Q2 regression for dollar raise with robust SE:
```

```r
coeftest(q2_lm_raise, vcov = vcovHC(q2_lm_raise, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                Estimate  Std. Error t value  Pr(>|t|)
## (Intercept) -685.789742  233.807443 -2.9331  0.003498 **
## sexF          25.860075   40.355920  0.6408  0.521924
## fieldOther   -89.409015   48.342644 -1.8495  0.064933 .
## fieldProf      1.249621   61.724909  0.0202  0.983855
## yrdeg         16.752126    3.920405  4.2731 2.278e-05 ***
## prev_salary   -0.001156    0.015454 -0.0748  0.940400
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
q2_lm_pct <- lm(
  raise_pct ~ sex + field + yrdeg + prev_salary,
  data = promotion_data
)
cat("\nQ2 regression for percent raise:\n")
```

```
##
## Q2 regression for percent raise:
```

```r
summary(q2_lm_pct)
```

```
##
## Call:
## lm(formula = raise_pct ~ sex + field + yrdeg + prev_salary, data = promotion_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.27621 -0.05158 -0.02007  0.02561  1.37505
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -4.893e-02  6.478e-02  -0.755   0.4504
## sexF         2.302e-03  1.208e-02   0.191   0.8489
## fieldOther  -2.593e-02  1.345e-02  -1.928   0.0543 .
## fieldProf   -1.316e-03  1.616e-02  -0.081   0.9351
## yrdeg        4.190e-03  1.024e-03   4.093 4.90e-05 ***
## prev_salary -2.828e-05  4.442e-06  -6.366 4.13e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1064 on 541 degrees of freedom
## Multiple R-squared:  0.07925,    Adjusted R-squared:  0.07074
## F-statistic: 9.313 on 5 and 541 DF,  p-value: 1.6e-08
```

```r
cat("\nQ2 regression for percent raise with robust SE:\n")
```

```
##
```

```
## Q2 regression for percent raise with robust SE:
```

```r
coeftest(q2_lm_pct, vcov = vcovHC(q2_lm_pct, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                Estimate  Std. Error t value  Pr(>|t|)
## (Intercept) -4.8927e-02  5.7037e-02 -0.8578   0.39138
## sexF         2.3017e-03  1.0238e-02  0.2248   0.82220
## fieldOther  -2.5934e-02  1.3012e-02 -1.9930   0.04676 *
## fieldProf   -1.3165e-03  1.6098e-02 -0.0818   0.93485
## yrdeg        4.1904e-03  9.8829e-04  4.2401 2.628e-05 ***
## prev_salary -2.8276e-05  4.7286e-06 -5.9799 4.059e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

5. Q3: Overall salary discrimination

```r
# avoid longitudinal methods by using one row per faculty (latest salary)
latest_salary <- salary_clean %>%
  arrange(id, year) %>%
  group_by(id) %>%
  slice_tail(n = 1) %>%
  ungroup() %>%
  mutate(
    log_salary = log(salary),
    experience = year - startyr
  )

latest_salary %>%
  summarise(
    n_people = n(),
    min_year = min(year, na.rm = TRUE),
    max_year = max(year, na.rm = TRUE)
  ) %>%
  print()
```

```
## # A tibble: 1 x 3
##   n_people min_year max_year
##      <int>    <int>    <int>
## 1     1597       95       95
```

```r
cat("\nLatest salary summaries by sex and rank:\n")
```

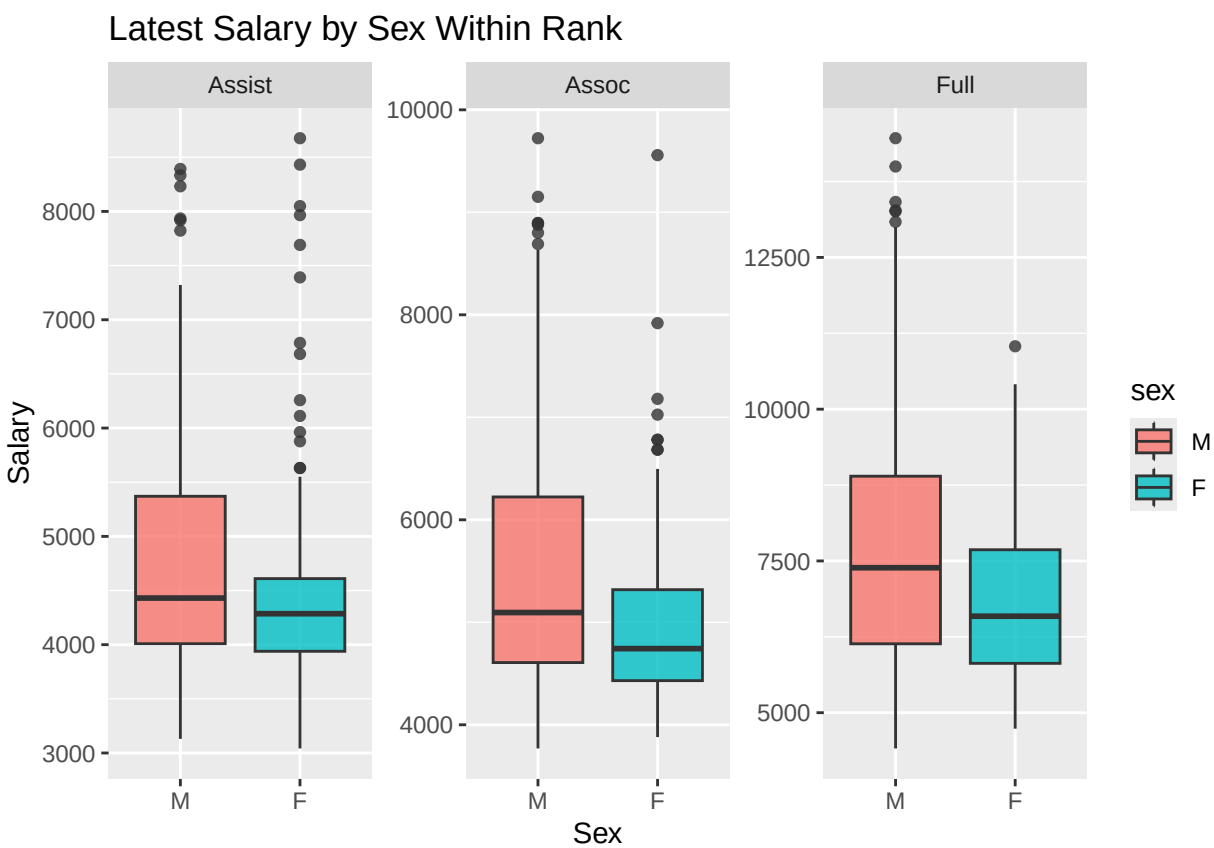
```
##
## Latest salary summaries by sex and rank:
```

```r
latest_salary %>%
  group_by(rank, sex) %>%
  summarise(
    n = n(),
    mean_salary = mean(salary, na.rm = TRUE),
    median_salary = median(salary, na.rm = TRUE),
    sd_salary = sd(salary, na.rm = TRUE),
    .groups = "drop"
```

```
  )
```

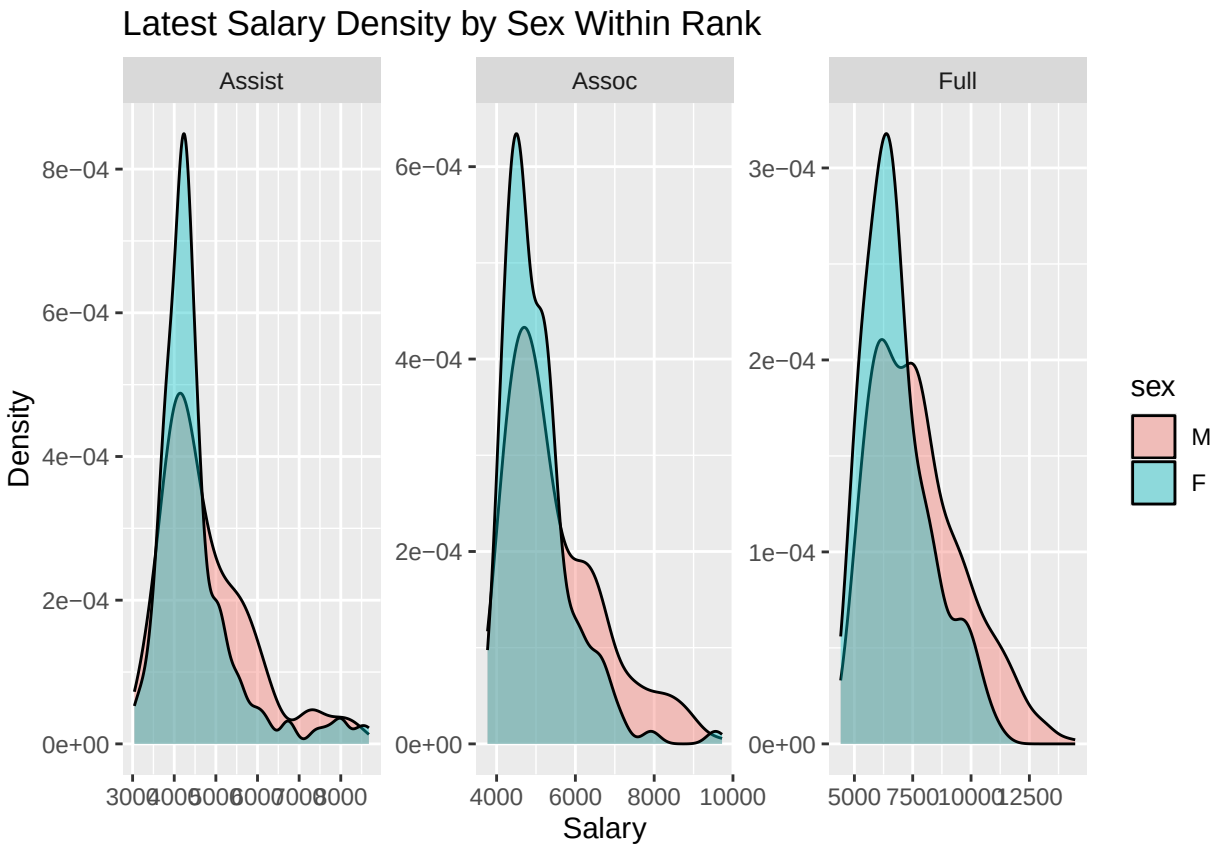
```
## # A tibble: 6 x 6
##   rank   sex       n mean_salary median_salary sd_salary
##   <fct>  <fct> <int>       <dbl>         <dbl>     <dbl>
## 1 Assist M       170       4774.         4430.     1117.
## 2 Assist F       145       4503.         4286       981.
## 3 Assoc  M       299       5480.         5095      1224.
## 4 Assoc  F       138       5019.         4743       858.
## 5 Full   M       719       7715.         7388      1943.
## 6 Full   F       126       6840.         6590      1436.
```

```
ggplot(latest_salary, aes(x = sex, y = salary, fill = sex)) +
  geom_boxplot(alpha = 0.8) +
  facet_wrap(~ rank, scales = "free_y") +
  labs(
    title = "Latest Salary by Sex Within Rank",
    x = "Sex",
    y = "Salary"
  )
```




```
ggplot(latest_salary, aes(x = salary, fill = sex)) +
  geom_density(alpha = 0.4) +
  facet_wrap(~ rank, scales = "free") +
  labs(
    title = "Latest Salary Density by Sex Within Rank",
    x = "Salary",
    y = "Density"
```

```
  )
```




```
q3_lm <- lm(
  salary ~ sex + rank + field + experience + admin,
  data = latest_salary
)
cat("\nQ3 overall salary model (latest salary):\n")
```

```
##
## Q3 overall salary model (latest salary):
```

```
summary(q3_lm)
```

```
##
## Call:
## lm(formula = salary ~ sex + rank + field + experience + admin,
##     data = latest_salary)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3334.3  -951.5  -164.3   600.3  6702.6
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 4016.626    131.041  30.652  < 2e-16 ***
## sexF        -409.325     87.059  -4.702 2.80e-06 ***
## rankAssoc    600.365    113.938   5.269 1.56e-07 ***
## rankFull    2607.643    122.202  21.339  < 2e-16 ***
```

```
## fieldOther   781.297    106.729   7.320 3.91e-13 ***
## fieldProf   1769.944    128.711  13.751  < 2e-16 ***
## experience    -4.236      4.603  -0.920    0.358
## admin1      1202.224    118.822  10.118  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1431 on 1589 degrees of freedom
## Multiple R-squared:  0.5084, Adjusted R-squared:  0.5062
## F-statistic: 234.7 on 7 and 1589 DF,  p-value: < 2.2e-16
```

```r
cat("\nQ3 overall salary model with robust SE:\n")
```

```
##
## Q3 overall salary model with robust SE:
```

```r
coeftest(q3_lm, vcov = vcovHC(q3_lm, type = "HC1"))
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value  Pr(>|t|)
## (Intercept) 4016.6261    88.3316 45.4721 < 2.2e-16 ***
## sexF        -409.3248    70.0167 -5.8461 6.095e-09 ***
## rankAssoc    600.3651    89.1285  6.7359 2.269e-11 ***
## rankFull    2607.6426   126.5920 20.5988 < 2.2e-16 ***
## fieldOther   781.2971    74.7331 10.4545 < 2.2e-16 ***
## fieldProf   1769.9444   108.0356 16.3830 < 2.2e-16 ***
## experience    -4.2364     5.6385 -0.7513    0.4526
## admin1      1202.2236   131.4548  9.1455 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# interaction model: does the salary gap differ by rank?
q3_interaction <- lm(
  salary ~ sex * rank + field + experience + admin,
  data = latest_salary
)
cat("\nQ3 salary model with sex x rank interaction:\n")
```

```
##
## Q3 salary model with sex x rank interaction:
```

```r
summary(q3_interaction)
```

```
##
## Call:
## lm(formula = salary ~ sex * rank + field + experience + admin,
##     data = latest_salary)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3374.8  -941.1  -187.9   594.1  6667.3
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3868.301    145.024  26.673  < 2e-16 ***
```

```
## sexF            -64.592    162.155  -0.398  0.69044
## rankAssoc       754.347    144.034   5.237 1.85e-07 ***
## rankFull       2817.331    143.439  19.641  < 2e-16 ***
## fieldOther      770.145    106.718   7.217 8.22e-13 ***
## fieldProf      1763.087    128.600  13.710  < 2e-16 ***
## experience       -4.811      4.598  -1.046  0.29565
## admin1         1205.407    118.575  10.166  < 2e-16 ***
## sexF:rankAssoc -314.264    218.627  -1.437  0.15079
## sexF:rankFull  -627.052    212.653  -2.949  0.00324 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1428 on 1587 degrees of freedom
## Multiple R-squared:  0.5111, Adjusted R-squared:  0.5083
## F-statistic: 184.3 on 9 and 1587 DF,  p-value: < 2.2e-16
```

```r
cat("\nQ3 interaction model with robust SE:\n")
```

```
##
## Q3 interaction model with robust SE:
```

```r
coeftest(q3_interaction, vcov = vcovHC(q3_interaction, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                 Estimate Std. Error t value  Pr(>|t|)
## (Intercept)    3868.3009    99.7946 38.7626 < 2.2e-16 ***
## sexF            -64.5917   111.5342 -0.5791 0.5625900
## rankAssoc       754.3475   114.2743  6.6012 5.545e-11 ***
## rankFull       2817.3313   144.8704 19.4472 < 2.2e-16 ***
## fieldOther      770.1445    74.4327 10.3469 < 2.2e-16 ***
## fieldProf      1763.0873   108.0410 16.3187 < 2.2e-16 ***
## experience       -4.8107     5.6345 -0.8538 0.3933496
## admin1         1205.4075   130.8587  9.2115 < 2.2e-16 ***
## sexF:rankAssoc -314.2637   145.4078 -2.1613 0.0308246 *
## sexF:rankFull  -627.0516   172.1399 -3.6427 0.0002785 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# log-salary
q3_lm_log <- lm(
  log_salary ~ sex + rank + field + experience + admin,
  data = latest_salary
)
cat("\nQ3 log-salary model:\n")
```

```
##
## Q3 log-salary model:
```

```r
summary(q3_lm_log)
```

```
##
## Call:
## lm(formula = log_salary ~ sex + rank + field + experience + admin,
##     data = latest_salary)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.48818 -0.14113 -0.01492  0.11271  0.68607
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.3226282  0.0182786 455.321  < 2e-16 ***
## sexF        -0.0590251  0.0121436  -4.861 1.29e-06 ***
## rankAssoc    0.1256876  0.0158930   7.908 4.84e-15 ***
## rankFull     0.4309180  0.0170457  25.280  < 2e-16 ***
## fieldOther   0.1200197  0.0148874   8.062 1.46e-15 ***
## fieldProf    0.2699668  0.0179535  15.037  < 2e-16 ***
## experience  -0.0003448  0.0006421  -0.537    0.591
## admin1       0.1670659  0.0165742  10.080  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1996 on 1589 degrees of freedom
## Multiple R-squared:  0.5678, Adjusted R-squared:  0.5659
## F-statistic: 298.2 on 7 and 1589 DF,  p-value: < 2.2e-16
```

```r
cat("\nQ3 log-salary model with robust SE:\n")
```

```
##
## Q3 log-salary model with robust SE:
```

```r
coeftest(q3_lm_log, vcov = vcovHC(q3_lm_log, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                Estimate  Std. Error  t value  Pr(>|t|)
## (Intercept)  8.32262821  0.01424517 584.2420 < 2.2e-16 ***
## sexF        -0.05902512  0.01079013  -5.4703 5.212e-08 ***
## rankAssoc    0.12568758  0.01485859   8.4589 < 2.2e-16 ***
## rankFull     0.43091799  0.01811909  23.7825 < 2.2e-16 ***
## fieldOther   0.12001965  0.01133042  10.5927 < 2.2e-16 ***
## fieldProf    0.26996679  0.01490157  18.1167 < 2.2e-16 ***
## experience  -0.00034480  0.00071706  -0.4808    0.6307
## admin1       0.16706593  0.01594811  10.4756 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# coefs
q3_log_coef <- coef(q3_lm_log)["sexF"]
cat("\nApproximate percent difference for women in latest salary:\n")
```

```
##
## Approximate percent difference for women in latest salary:
```

```r
cat(round((exp(q3_log_coef) - 1) * 100, 2), "%\n")
```

```
## -5.73 %
```